

Email Generation Assistant

AI Engineer Candidate Assessment - Final Report

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Section 1 - Project Overview

This project is a working prototype that takes three inputs - Intent, Key Facts, and Tone - and generates a polished, ready-to-send professional email using the Gemini API. A Streamlit web interface (app.py) provides the user-facing experience.

1.1 How to Run

- Install dependencies: `pip install -r requirements.txt`
- Set API key: `set GEMINI_API_KEY=your_key_here`
- Launch UI: `streamlit run app.py`
- Evaluation: `python evaluate.py`
- Comparison: `python comparison.py`

Section 2 - Advanced Prompt Engineering

The assistant uses Role-Playing + Chain-of-Thought (CoT) - two complementary techniques combined deliberately, each targeting a different failure mode.

2.1 Technique Breakdown

Technique	What it does	Failure mode it targets
Role-Playing	Gives the model the persona of an experienced corporate communication specialist.	Generic, robotic output that sounds like a template.
Chain-of-Thought (CoT)	Four numbered steps force the model to reason about intent, facts, and tone before writing prose.	Dropped facts and tone-inconsistent openings/closings.

2.2 Prompt Template

```
You are an experienced corporate communication specialist.  
Follow these steps:  
1. Understand the user's intent.  
2. Include all key facts naturally.  
3. Adjust the writing style according to the tone.  
4. Produce a professional email with a subject line.  
  
Intent: {intent}  
Key Facts: {key_facts}  
Tone: {tone}  
  
Generate the final email.
```

Section 3 - Custom Evaluation Metrics

Three custom metrics were designed specifically for email generation, covering completeness, appropriateness, and readability.

Metric 1 - Fact Recall Score (0-100)

Definition: Measures whether every key fact provided by the user appears in the generated email.

Logic: For each fact, significant keywords (ignoring stop words) are extracted. A fact is counted as recalled if at least 50% of its keywords appear in the generated email. $\text{Score} = (\text{facts recalled} / \text{total facts}) \times 100$.

Method: Pure Python keyword matching - deterministic, no API call.

Metric 2 - Tone Accuracy Score (0-100)

Definition: Measures how well the generated email's vocabulary, sentence structure, and overall feel match the requested tone.

Logic: LLM-as-a-Judge - a separate Gemini call rates 0-100 how well the email matches the target tone. Output is constrained to a single integer for reliable parsing.

Method: Gemini API (LLM-as-a-Judge).

Metric 3 - Conciseness Score (0-100)

Definition: Measures whether the email is an appropriate, readable length.

Logic: Emails with 60-150 words score 100. Outside that band, score drops proportionally: -2/word below 60, -1.5/word above 150 (floored at 0).

Method: Pure Python word-count check.

Section 4 - Test Scenarios

Ten unique (Intent, Key Facts, Tone) scenarios were created, each paired with a human-written reference email. All 10 are stored in scenarios.json.

#	Intent	Tone
1	Follow up after meeting	Formal
2	Request for proposal details	Formal
3	Apology for delayed delivery	Empathetic
4	Notify team of server outage	Urgent
5	Thank you note after a job interview	Formal
6	Payment reminder to client	Formal
7	Product launch announcement	Casual
8	Reschedule a meeting	Casual
9	Respond to a customer complaint	Empathetic
10	Renew a partnership agreement	Formal

Section 5 - Evaluation Results (Part 2)

All 10 scenarios were run through Model B (Role-Play + CoT) via evaluate.py, producing results.csv. The overall average score across all 3 custom metrics was 85.00 / 100.

Scenario	Intent	Tone	Fact Recall	Tone Accuracy	Conciseness	Average
1	Follow up after meeting	Formal	100.0	95.0	94.0	96.33
2	Request for proposal details	Formal	100.0	98.0	25.0	74.33

3	Apology for delayed delivery	Empathetic	100.0	98.0	29.5	75.83
4	Notify team of server outage	Urgent	100.0	95.0	65.5	86.83
5	Thank you note after a job interview	Formal	100.0	98.0	94.0	97.33
6	Payment reminder to client	Formal	100.0	100.0	100.0	100.0
7	Product launch announcement	Casual	100.0	95.0	67.0	87.33
8	Reschedule a meeting	Casual	100.0	92.0	100.0	97.33
9	Respond to a customer complaint	Empathetic	100.0	95.0	5.5	66.83
10	Renew a partnership agreement	Formal	100.0	98.0	5.5	67.83
	Overall Average		100.00	96.40	58.60	85.00

Observation: Fact Recall was perfect or near-perfect across every scenario (avg 100.0), and Tone Accuracy stayed consistently high (avg 96.4). Conciseness was the most variable metric (avg 58.6), occasionally dropping sharply when the model produced emails well over the 150-word target - this is the main quality lever for future prompt refinement.

Section 6 - Model Comparison Analysis (Part 3)

Both strategies were run across the same 10 scenarios using the same 3 custom metrics via comparison.py, producing comparison.csv (20 rows).

Strategies Compared

Strategy	Model	Prompt
Model A (Basic Prompt)	gemini-2.5-flash-lite	Plain instruction with no role or reasoning steps.
Model B (Role-Play+CoT)	gemini-2.5-flash-lite	Role-Playing + Chain-of-Thought (see Section 2).

Aggregate Scores (10 scenarios each)

Strategy	Fact Recall	Tone Accuracy	Conciseness	Overall Avg
Model A (Basic Prompt)	93.33	91.00	73.55	85.96
sModel B (Role-Play + CoT)	93.33	91.50	79.35	88.06

Full Comparison Table

Scenario	Strategy	Fact	Tone	Concise	Avg
1	A (Basic)	100.0	95.0	79.0	91.33
	B (Role+CoT)	100.0	95.0	31.0	75.33
2	A (Basic)	100.0	95.0	82.0	92.33
	B (Role+CoT)	100.0	95.0	100.0	98.33
3	A (Basic)	66.67	90.0	0	52.22
	B (Role+CoT)	100.0	90.0	0	63.33
4	A (Basic)	100.0	90.0	100.0	96.67
	B (Role+CoT)	100.0	95.0	100.0	98.33
5	A (Basic)	100.0	95.0	100.0	98.33
	B (Role+CoT)	100.0	95.0	100.0	98.33
6	A (Basic)	100.0	95.0	100.0	98.33
	B (Role+CoT)	100.0	95.0	100.0	98.33
7	A (Basic)	100.0	85.0	89.5	91.5
	B (Role+CoT)	100.0	85.0	82.0	89.0
8	A (Basic)	66.67	85.0	100.0	83.89
	B (Role+CoT)	66.67	85.0	100.0	83.89
9	A (Basic)	100.0	85.0	85.0	90.0
	B (Role+CoT)	100.0	85.0	100.0	95.0
10	A (Basic)	100.0	95.0	0	65.0

	B (Role+CoT)	66.67	95.0	80.5	80.72
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Comparative Analysis

1. Which model/strategy performed better?

Model B (Role-Play + CoT) performed better overall, scoring 88.06/100 on average versus 85.96/100 for Model A (Basic Prompt) — a margin of 2.10 points across the same 10 scenarios. Fact Recall was nearly identical between the two strategies (93.33 for Model A vs 93.33 for Model B), and Tone Accuracy was also close (91.00 vs 91.50). The decisive metric was Conciseness, where Model B averaged 79.35 versus Model A's 73.55.

2. Biggest failure mode of the lower-performing model (Model A (Basic Prompt))?

Model A (Basic Prompt)'s weakest metric was Conciseness (avg 73.55), well below its other two metrics. The clearest concrete example is Scenario 3 ("Apology for delayed delivery"), where Model A (Basic Prompt) produced a 232-word email — well past the 150-word ceiling defined for professional emails in this project — scoring 0 on Conciseness for that scenario. Without an explicit reasoning step forcing the model to plan structure before writing, the basic prompt has no mechanism to keep output length in check, so it defaults to an over-elaborated, padded structure in scenarios that call for brevity.

3. Which model do you recommend for production, and why?

Model B (Role-Play + CoT) is recommended for production. With Fact Recall and Tone Accuracy effectively tied between the two strategies, the 2.10-point overall advantage comes almost entirely from more consistent Conciseness control (79.35 vs 73.55 avg) — the Chain-of-Thought step, which explicitly asks the model to plan an opening, fact order, and closing before drafting, appears to curb the open-ended elaboration seen in the basic prompt. This makes Model B (Role-Play + CoT) the safer default for an email assistant where length and professionalism both matter. Caveat: the comparison is not unanimous at the scenario level — in Scenario 1 ("Follow up after meeting"), Model A actually scored 16 points higher than Model B because Model B's CoT reasoning produced a noticeably longer email in that case. This suggests Conciseness is the primary axis worth further prompt tuning regardless of which strategy is deployed (e.g., adding an explicit word-count instruction to the prompt template) rather than treating either strategy as a finished, perfectly tuned solution.